

Green color font is the syllabus for 1st CT. Red color font is to be taught. Black color font is already taught but not in CT syllabus. Blue is for instruction

Fundamental electrical concepts: Different measuring units

**DC voltage
current**

**resistance and power; Series
networks definitions
mesh and node circuit analysis**

**reduction of a complicated
network**

**conversion between T and π
section.**

Networks transformations:

Equivalent circuit

Superposition theorem

the reciprocity theorem

Thevenin's theorem

Norton's theorem

**Maximum power transfer
theorem.**

Introduction to magnetic Circuit:
Transformer working principle
construction
and maintenance
transformer's e.m.f equations

Transformer regulation and
efficiency
different types of transformer.

Electromagnetic Field:
Maxwell's Equation
Electromagnetic wave equation
and propagation
Pointing vector

Faraday's laws of electromagnetic induction

Lenz's Law

Motional e.m.f

Eddy current. Self and Mutual Inductance.

Capacitor and Capacitance:

Capacitance

Parallel plate capacitor

cylindrical capacitor

spherical capacitor

capacitors in series & parallel

energy stored in a capacitor

transformers.

**A.C Circuits: Instantaneous and
r.m.s. values of current
voltage and average power**